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DIVISION - C

Title: PCB Recycling in Goa — Integration of Bioleaching (Biometallurgy)

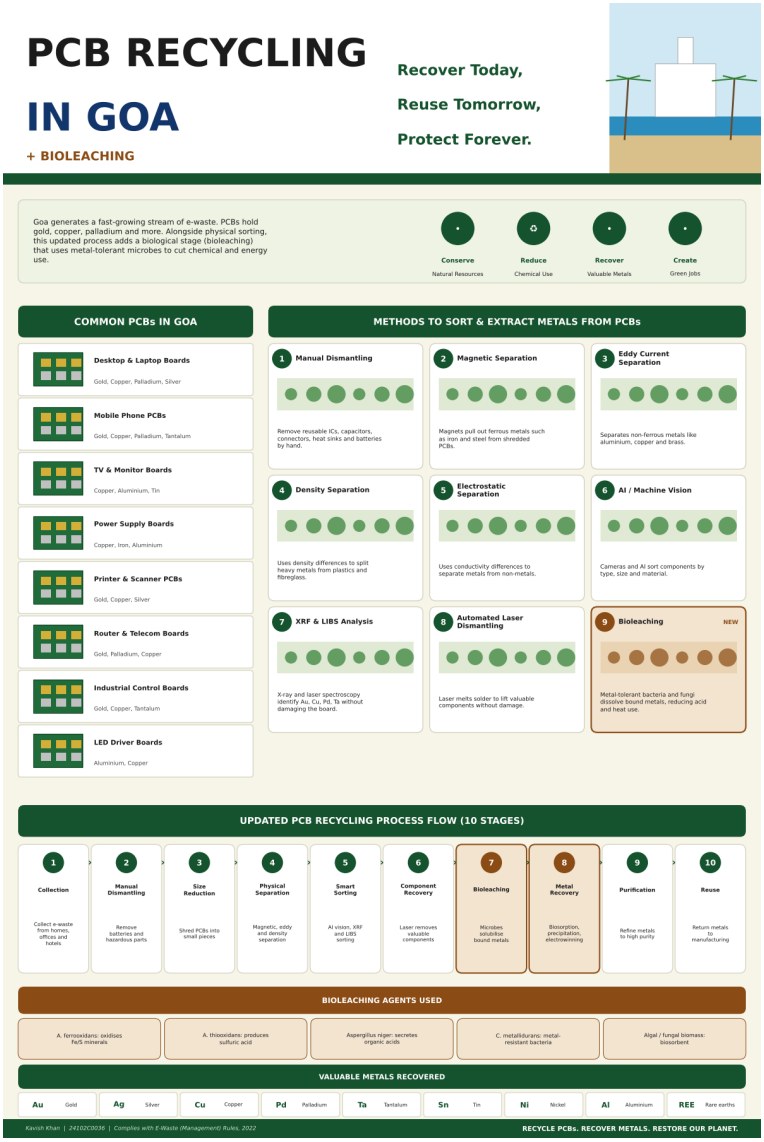


Figure 1: The updated PCB recycling poster for Goa

Figure 1 shows the PCB recycling process for Goa after a biological stage was added. The original poster relied on mechanical sorting followed by chemical (hydro/pyro) metal extraction. The aim of this update is to bring in bioleaching, where metal-tolerant bacteria and fungi dissolve the metals bound in shredded PCB material, so that less acid, heat and energy is needed in the later extraction steps. As the figure shows, the recycling flow now runs through ten stages instead of nine: collection, manual dismantling, size reduction, physical separation, smart sorting, component recovery, bioleaching, metal recovery, purification, and reuse. The biological stages are highlighted in brown so that they stand apart from the green mechanical and chemical stages.

Changes made:

Added a new method card — Bioleaching — to the metal sorting and extraction methods section, explaining that metal-tolerant microbes dissolve bound metals.

Added a new stage, Bioleaching, to the process flow after component recovery, and replaced the old hydro/pyro/bio-leaching extraction step with a separate Metal Recovery stage that uses biosorption, precipitation and electrowinning.

Added a new section listing 5 bioleaching agents and their function:

Acidithiobacillus ferrooxidans — oxidises Fe/S minerals

Acidithiobacillus thiooxidans — produces sulfuric acid for leaching

Aspergillus niger — fungal leacher, secretes organic acids

Cupriavidus metallidurans — metal-resistant bacteria

Algal / fungal biomass — biosorbent used in the recovery stage

Merged the separate XRF and LIBS methods into a single analysis card to make room for the new bioleaching card.

Changed one of the four headline benefits from “Reduce Environmental Pollution” to “Reduce Chemical Use”, and added a note in the introduction about lower chemical and energy consumption.

Added a Valuable Metals Recovered strip (Au, Ag, Cu, Pd, Ta, Sn, Ni, Al and rare earths) and redesigned the layout with a green theme and a brown accent for the biological steps.